Statistical Models

TUHH, Summer Term 2024

Exercise Sheet 3 (Ay, June 27, 2024)

Exercise 3.1 Consider the following map which we refer to as a translation:

$$+ : \mathcal{P}_+(I) \times (\mathcal{F}(I)/\mathbb{R}) \to \mathcal{P}_+(I),$$

$$(\mu, [f]) \mapsto \mu + [f] := \frac{e^f}{\mathbb{E}_\mu(e^f)}\mu = \sum_{i \in I} \frac{e^{f(i)}}{\mathbb{E}_\mu(e^f)} \mu_i \, \delta^i,$$

where $[f]$ denotes the equivalence class $\{f + g \,:\, g \text{ is a constant function}\}$. Show that this translation is well-defined and satisfies the following conditions so that $\mathcal{P}_+(I)$ is an affine space over $\mathcal{F}(I)/\mathbb{R}$:

(i) $\mu + [0] = \mu$

(ii) $(\mu + [f]) + [g] = \mu + ([f] + [g])$

(iii) For all $\mu, \nu \in \mathcal{P}_+(I)$ there exists a unique $[f] \in \mathcal{F}(I)/\mathbb{R}$ satisfying $\mu + [f] = \nu$.

Exercise 3.2 Consider two binary variables X_1, X_2 with values in $\{0, 1\}$ and joint distribution μ on $\{0, 1\} \times \{0, 1\} = \{(0, 0), (1, 0), (0, 1), (1, 1)\} =: I$, that is

$$\mu = \mu_{00}\, \delta^{00} + \mu_{10}\, \delta^{10} + \mu_{01}\, \delta^{01} + \mu_{11}\, \delta^{11}.$$

The marginals of any probability measure $\nu \in \mathcal{P}(I)$ are given by

$$\nu^{(1)} = \nu_0^{(1)}\, \delta^0 + \nu_1^{(1)}\, \delta^1 = (\nu_{00} + \nu_{01})\, \delta^0 + (\nu_{10} + \nu_{11})\, \delta^1,$$

$$\nu^{(2)} = \nu_0^{(2)}\, \delta^0 + \nu_1^{(2)}\, \delta^1 = (\nu_{00} + \nu_{10})\, \delta^0 + (\nu_{01} + \nu_{11})\, \delta^1.$$

Find among those probability measures ν that have the same marginals as μ the one with maximal entropy.

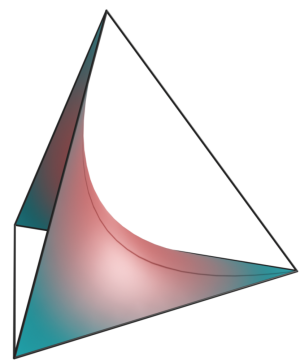
Exercise 3.3 This exercise continues with the context of Exercise 3.2. Consider the two functions

$$f_j : \{0, 1\} \times \{0, 1\} \to \mathbb{R}, \quad (x_1, x_2) \mapsto \begin{cases} 1, & \text{if } x_j = 1 \\ 0, & \text{otherwise} \end{cases}, \quad j = 1, 2.$$

These functions define the exponential family

$$\mathcal{E} \;:=\; \left\{ \frac{1}{Z(\theta_1,\theta_2)} \exp\left(\theta_1 f_1 + \theta_2 f_2\right) \;:\; \theta_1,\theta_2 \in \mathbb{R} \right\},$$

which is referred to as the *independence model*. This model is illustrated in the following figure (by Tobias Heidorn; the colours have no special meaning here):



Given a probability measure $\mu \in \mathcal{P}_+(I)$, find $\mu^* \in \mathcal{E}$ that satisfies

$$D_{KL}(\mu\|\mu^*) \;=\; \min_{\nu\in\mathcal{E}} D_{KL}(\mu\|\nu).$$

Compare μ^* with the maximum entropy solution of Exercise 3.2.

Exercise 3.4 We consider again the same setting as Exercise 3.2. We let X_1 and X_2 be independent and parametrise their distributions using a single parameter θ in the following way:

$$P(X_1 = 1) = P(X_2 = 1) = \theta. \tag{1}$$

Note that this model is a 1-dimensional submanifold of the independence model. We now consider the following two statistics:

$$f_1(x_1,x_2) = x_1 + x_2, \tag{2}$$
$$f_2(x_1,x_2) = x_1. \tag{3}$$

Show that $G(\theta) = G_{f_1}(\theta) > G_{f_2}(\theta)$.

This implies that the statistic f_1 is sufficient and f_2 is not, and therefore $f_1(x_1, x_2)$ contains all the information about the parameter θ contained in the outcome (x_1, x_2) but $f_2(x_1, x_2)$ does not.

Exercise 3.5 Consider $I = \{1, \ldots, n\}$, a matrix $A = (a_{ij})_{i,j \in I}$, and the following vector field on $\mathcal{P}_+(I)$:

$$V : \mathcal{P}_+(I) \to \mathcal{S}_0(I), \quad \mu \mapsto V(\mu) = \sum_{i \in I} V_i(\mu)\, \delta^i,$$

$$V_i(\mu) = \mu_i \left(\sum_{j \in I} a_{ij} \mu_j - \sum_{k \in I} \mu_k \sum_{j \in I} a_{kj} \mu_j \right), \quad i \in I.$$

Show that this vector field is a Fisher-Rao gradient field whenever the matrix A is symmetric. Provide a corresponding potential function for that case.